

ARANI ROY

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EDUCATION

Purdue University

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Kaushik Roy

Aug 2021 – Dec 2026 (Expected)

GPA: 3.8/4.0

West Lafayette, IN

Jadavpur University

B.E. in Electronics and Telecommunication Engineering

Kolkata, India

2014 – 2018

GPA: 9.35/10.0

RESEARCH SUMMARY

Ph.D. researcher working on generative AI across multimodal and language models (diffusion models, diffusion language models, and LLMs), representation geometry, model unlearning, efficient reasoning, and efficient deep learning. My research focuses on developing training-free and geometry-aware methods for controlling, analyzing, compressing, and selectively modifying foundation models, along with faster reasoning, backpropagation-free learning, and compression-aware training for efficient AI systems.

SELECTED PUBLICATIONS

- A. Roy, S.D. Biswas, K. Roy. “HEART: Hyperspherical Embedding Alignment via Kent-Representation Traversal in Diffusion Models.” [NeurIPS 2026](#).
- A. Roy*, S.D. Biswas*, K. Roy. “CURE: Concept Unlearning via Orthogonal Representation Editing in Diffusion Models.” [NeurIPS 2025, Spotlight](#).
- A. Roy, M.P.E. Apolinario, S.D. Biswas, K. Roy. “Feedback Alignment Meets Low-Rank Manifolds: A Structured Recipe for Local Learning.” [WACV 2026, Oral, Best Student Paper – Algorithms, CVPRW 2025](#).
- A. Roy*, S.D. Biswas*, K. Roy. “SlimDiff: Training-Free, Activation-Guided Slimming of Diffusion Models.” *Under Review*.
- S.D. Biswas, A. Roy, K. Roy. “Now You See It, Now You Don’t – Instant Concept Erasure for Safe Text-to-Image and Video Generation.” [CVPR Findings 2026](#).
- M.P.E. Apolinario, A. Roy, K. Roy. “LLS: Local Learning Rule for Deep Neural Networks Inspired by Neural Activity Synchronization.” [WACV 2025](#).
- S.D. Biswas, E. Soufleri, A. Roy, K. Roy. “Towards Scalable Modeling of Compressed Videos for Efficient Action Recognition.” [WACV 2026](#).
- A. Roy, K. Roy. “AlphaBlend: Hardware-Algorithm Co-design with Mixed-Alphabet Set Multipliers for DNN Workloads.” [ISCAS 2025](#).

* Equal contribution.

RESEARCH EXPERIENCE

C-BRIC/ NRL Lab, Purdue University

Graduate Research Assistant/ Fellow

West Lafayette, IN

Aug 2021 – Present

HEART: Geometry-Aware Controllable Diffusion Generation

2026

- Developed a **training-free, geometry-aware framework** for controllable diffusion generation by modeling text embeddings on their intrinsic hyperspherical geometry and replacing conventional Euclidean semantic edits with **Kent-representation traversal**.
- Enabled precise **subject and attribute edits** across UNet and DiT models with strong scene preservation.
- Achieved the **best BG-/FG-LPIPS trade-off** and **lowest application overhead** among compared methods. Published at [NeurIPS 2026](#).

CURE: Training-Free Concept Unlearning in Diffusion Models

2024 – 2025

- Developed a fast, **training-free concept-unlearning framework** that performs **closed-form cross-attention weight editing** using orthogonal projection and spectral representation geometry.

- Performed **closed-form text-attention editing** for concept removal, improving erasure by up to **16%** while preserving image quality and adversarial robustness. Published at **NeurIPS 2025 (Spotlight)**.

SlimDiff: Training-Free Diffusion Model Compression

2024 – 2025

- Developed an **activation-guided, timestep-aware** diffusion compression framework using operator-aware low-rank decomposition.
- Achieved **35% faster inference** and **~100M parameter reduction** while preserving generation quality.
- Designed **SlimSet**, a 500-sample calibration set, and spectral influence criteria for activation-weight and timestep-aware profiling, reducing calibration cost by **98%**.

Backpropagation-Free Learning via Structured Low-Rank Geometry

2023 – 2025

- Developed a **backpropagation-free local-learning framework** combining Direct Feedback Alignment with low-rank manifold constraints.
- Used orthogonality-preserving updates on low-rank manifolds to improve representation alignment and achieve performance competitive with backpropagation while maintaining low inference overhead.
- Scaled to **32-layer ResNets** and **ImageNet-scale settings**. Published at **CVPRW 2025** and **WACV 2026 (Oral)**, receiving the **Best Student Paper – Algorithms Award**.

Hardware-Algorithm Co-design for Efficient Neural Networks

2022 – 2023

- Developed a non-uniform, **approximate-multiplier-aware quantization framework** co-optimized with the target accelerator architecture.
- Demonstrated approximately **3.5× energy savings** and **1.3× speedup** while maintaining model accuracy. Published at **ISCAS 2025**.

Ongoing Projects

- **ARC: Aligned Riemannian Concept Erasure in Diffusion Models**

Developing a **geometry-aware concept and feature unlearning framework** that models diffusion-model text representations on a hyperspherical manifold and derives **geodesic projection operators** for localized, training-free concept and feature removal while preserving neighboring semantics.

- **Feature and Concept Unlearning in Large Language Models**

Analyzing and disentangling **LLM representation spaces** to understand how semantic features, concepts, and task-specific knowledge are encoded across transformer layers, with the goal of enabling **concept-level, and sequential-task unlearning** while preserving unrelated model capabilities.

- **Semantic Reconstruction and Parallel Reasoning in Diffusion Language Models**

Developing a **position-relaxed reconstruction objective** for masked diffusion language models to improve deep-mask semantic reconstruction and enable more effective **parallel decoding** on language and reasoning tasks.

- **Model- and Task-Aware Data Curation for Multimodal Learning**

Developing data curation methods that select samples with respect to the **modality gap**, improving cross-modal alignment and data efficiency for vision-language models.

INDUSTRY EXPERIENCE

Texas Instruments

May 2023 – Aug 2023

Compute-in-Flash Research Intern

Dallas, TX

- Prototyped a **digital compute-in-flash architecture** for neural-network matrix-vector workloads and ported a 65 nm design to **28 nm flash**, analyzing energy-throughput trade-offs.

Samsung Semiconductor India R&D

Jul 2018 – Jul 2021

Associate Staff Engineer

Bengaluru, India

- Led library-design projects across **4 nm–130 nm nodes**, automating design flows for **~2× faster turnaround**. Designed PPA-optimized sequential circuits resulting in **three patents** and the **Samsung President's Award**.

Samsung Electronics

May 2017 – Jul 2017

Student Trainee

Bengaluru, India

- Developed a synthesizable **memory compiler** with an **RTL-to-GDS flow** resulting in **Best Intern Project** award.

PATENTS

- **Three Samsung patents** on performance- and power-optimized flip-flop architectures for advanced standard-cell libraries.

HONORS & AWARDS

- **Best Student Paper - Algorithms**, WACV 2026, **Oral Presentation**, WACV 2026, **Spotlight Paper**, NeurIPS 2025
- **Bilsland Dissertation Fellowship**, Purdue University, 2026
- **Frederick N. Andrews Fellowship**, Purdue University
- **Samsung President’s Award**, innovation in 4 nm / 5 nm circuit design
- **Samsung Best Intern Project Award**
- **Best Paper – User Design Track**, 33rd International Conference on VLSI Design & Embedded Systems
- **Sanghamitra Gold Medal**, Jadavpur University

TECHNICAL SKILLS

Research	Diffusion Models, Large Language Models, Multimodal Foundation Models, Model Editing and Unlearning, Representation Geometry, Model Compression, Efficient Deep Learning
ML / AI	PyTorch, Hugging Face Transformers, Diffusers, CUDA, NumPy, Scikit-learn
Programming	Python (proficient), C++, Bash, Java
Hardware / EDA	Verilog, HSPICE, Cadence Virtuoso, Synopsys ICC2, FineSIM

ACADEMIC SERVICE

Reviewer Service: NeurIPS 2026, ICML 2026, ECCV 2026, CVPR 2026, ICLR 2026, NeurIPS 2025, ICLR 2025, WACV 2026, ICCV 2025.